Gregorio Zaltzman D’ Ambrosio

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Education

University of California, San Diego, Bachelor of Science in Aerospace Engineering, GPA: 3.2 *(Sep 2022 - Present)*

Relevant Coursework: Fluid Dynamics, Solid Mechanics, Statics & Dynamics, Thermodynamics, Heat exchange, Ordinary Differential Equations, Numerical Methods, Aerospace Structures, Propulsion, Signals and Systems, Linear Control, Aerodynamics, Aerospace Controls, Orbital Mechanics, Engineering Experimental Techniques

British School of Paris A-levels: *A** Math, *A* Physics, *A* Drama, *A** EPQ, GCSE Scholar, SAT: M:780 E: 660 Dec 2022
Duke of Edinburgh - Gold Award, Academic Prefect, Red Cross Volunteer

Skills

Programming: MATLAB, Python, Arduino IDE
Design: SolidWorks, SolidWorks FLOW, Fusion 360, STK, FEA, XFLR5, Open VSP, AVL, Microsoft Office

Languages: English, Spanish, French (All fluent)
Soft Skills: Collaboration, Problem-solving, Teamwork, Analytical Thinking, Communication

Projects

Conceptual Design – Blended Wing Body (Chief Engineer)
Designed an ultra-fuel-efficient Blended Wing Body commercial aircraft. focusing on aerodynamics, design optimization, unconventional configurations, and direct operating cost analysis; developed a full CAD model and 3D render while performing structural layout, propulsion integration, and static/dynamic stability analyses.

Wing Spar Analysis Project
Developed a MATLAB program to analyze single-cell wing bending, torsion, and shear for a 4-stringer wing with uniform skin under concentrated and distributed loads.

SeaGlide Build Project
Manufacturing, assembly, and testing of an autonomous buoyancy-driven underwater glider, contributing to structural fabrication, electronics integration, and system validation. Soldering electronic components, Arduino setup and programming for control systems, buoyancy engine assembly and calibration, and subsystem testing to ensure proper vehicle operation.

Low-Speed Aircraft Aerodynamics Workflow (XFLR5, OpenVSP, AVL)
Executed an end-to-end aerodynamics workflow using XFLR5, OpenVSP, and AVL to perform airfoil selection, conceptual aircraft modeling, and static stability analysis.

Work Experience

DBF @ UC San Diego - *Aerodynamics & Structures* subteams — San Diego, September 2025 - Present
Worked on RC competition aircraft. Contributing to airfoil selection, configuration optimization, structural sizing, and performance-driven design trade studies, engineering analysis, CAD modeling, and interdisciplinary teamwork.

RPL @ UC San Diego - *Daedalus Rocket* — San Diego, June 2025 - Present
Supported the design, build, and launch of a G-class model rocket (~2,100 ft), contributing to assembly, CAD support, testing preparation, and team design discussions while gaining exposure to rocket stability, avionics integration, and contributed to flight performance analysis in MATLAB.

UC San Diego ITS - *Service Desk Lead Technician* — San Diego, September 2023 - Present
Led a team of 60 + technicians by supporting and maintaining a positive, collaborative work environment. Oversee daily Service Desk operations, including complex technical support, workflow coordination, upkeeping documentation, and extensive communication with other departments. Also contributed to long-term team growth through training, hiring support, project work, and improving processes and resources for the Service Desk.

C2N @ CNRS - *Shadow Intern* — Paris, France, August 2022
Analysed nanophotonic crystals and nanolasers made using nanocavities, learned about experimental procedures in a clean room, such as the creation of nanolasers, and the operation of an electron microscope. Networked with a quantum computing startup that was developing and manufacturing quantum computers

Leadership

Phi Gamma Delta at UC San Diego *– Scholarship, Graduate Relations, Brotherhood chair*
Led professional brotherhood events focused on leadership development, teamwork, and community engagement, contributing to event organization and collaborative initiatives while developing interpersonal and organizational skills. Ran mentorship programs with younger engineers and paired up other new members to receive valuable mentorship from older members.

International Student Society— *Event Coordinator & Recruitment Chair*
Mentored 50+ international students, organized community events, and revamped the club membership pipeline

Duke of Edinburgh Gold Award – *British School of Paris*
Completed the Duke of Edinburgh Gold Award, demonstrating sustained leadership, resilience, and teamwork